UEmbed: Unified Sparse and Dense Multimodal Embeddings

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Abstract

Sparse retrieval underpins modern search systems, from web search to retrieval-augmented generation. Existing work has introduced Learned Sparse Retrieval (LSR) to push beyond exact lexical matching toward richer semantics. Yet LSR has so far remained tied to encoder-style bidirectional architectures, and its extension to multimodal settings still relies heavily on auxiliary cross-modal modules. To address these limitations, we introduce UEMBED (Unified Embedding), a decoder-only multimodal embedding model that produces both sparse lexical and dense representations in one causal forward pass. UEMBED appends N learnable special tokens to the input and partitions the vocabulary into N disjoint subsets. Each token's causal hidden state predicts sparse weights over its assigned subset, and the N subsets are concatenated into the full sparse vector. Trained on public data, we release UEMBED at 2B, 4B, and 9B scales. UEMBED-9B reaches 71.8 (dense) and 71.0 (sparse) on MMEB-v2, outperforming existing off-the-shelf multimodal embedding models (*e.g.,* RzenEmbed). On BEIR, UEMBED also remains competitive with strong dense and sparse baselines. Furthermore, we demonstrate the practical utility of UEMBED across three dimensions: effectiveness, efficiency, and agentic applications. Overall, UEMBED offers a new paradigm for retrieval: it unifies dense and sparse embeddings in one model, while further extending sparse retrieval to unify text and multimodal inputs.

1 Introduction

Information retrieval (IR) underpins a broad range of real-world applications (Zhang et al., 2022; Wei et al., 2026; Zhang et al., 2026) , including web search and question answering. Sparse lexical methods such as BM25 (Robertson and Zaragoza, 2009) remain widely deployed for their efficiency and interpretability, but their reliance on exact term matching limits their ability to capture semantic similarity beyond surface-level lexical overlap. Learned sparse retrieval (LSR) addresses this gap by training neural models to produce sparse lexical representations. Representative methods such as SPLADE (Formal et al., 2021b,a, 2022) combine token-level transformations with max pooling, substantially improving retrieval quality over classical term-weighting schemes.

Despite this progress, LSR remains limited in three key respects. (1) *Architectural constraint:* current LSR methods are largely built on encoder-style, bidirectional architectures (*e.g.,* BERT (Devlin et al., 2019)). Due to the causal masking of unidirectional attention, decoder-only models cannot directly reuse traditional pooling strategies to capture the global context needed for sparse representations. Leveraging large language models (LLMs) for sparse retrieval has typically required either converting the causal backbone to a bidirectional one (Chen et al., 2024) or adopting dedicated training curricula (Xu et al., 2026), both of which incur additional training cost and break compatibility with optimized causal-model serving frameworks such as vLLM (Kwon et al., 2023). (2) *Limited modality:* most LSR methods remain confined to text, as traditional approaches rely heavily on text-only BERT architectures that lack native multimodal capabilities. The few multimodal methods (Lu et al., 2021; Chen et al., 2023) introduce auxiliary cross-modal modules rather than deriving sparse representations directly from a single multimodal backbone, making them difficult to extend to new modalities. (3) *Unexplored practical utility:* existing sparse methods are predominantly evaluated on traditional benchmarks such as COCO (Lin et al., 2014), leaving the practical advantages of sparse retrieval under-examined beyond accuracy.

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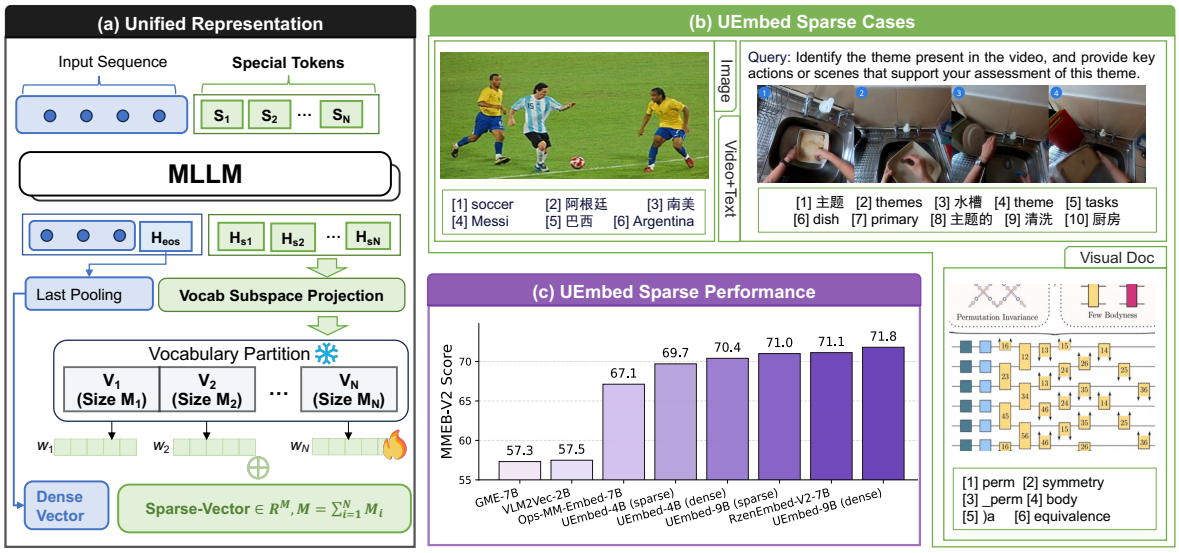


Figure 1: Overview of UEMBED. (a) Framework: we append N learnable special tokens to a decoder-only MLLM and assign each token a disjoint vocabulary subset. (b) Sparse cases: examples of activated lexical tokens produced by UEMBED-9B. (c) Performance: UEMBED achieves strong sparse and dense retrieval results on MMEB-v2.

The primary challenge in deriving sparse representations from causal models is the information bottleneck: relying on a single token (*e.g.,* EOS, CLS) to project into a massive $|V|$-dimensional vocabulary space severely limits representational capacity. To overcome this, we present UEMBED (**U**nified **Embed**ding), which appending N learnable special tokens to a MLLM and assigns each token a disjoint subset of the vocabulary. Specifically, these subsets are derived via k-means clustering, which explicitly encourages each token to represent a distinct spatial direction. Under causal attention, each special token summarizes the input from its position and predicts sparse weights over its assigned cluster. The N subsets are then concatenated into the full sparse vector. This partitioned projection effectively circumvents the single-token representational bottleneck. The dense embedding is obtained by last-pooling the EOS token preceding the special tokens. As shown in Figure 1(b), UEMBED produces semantically meaningful lexical activations, indicating that the sparse representation captures cross-modal semantics.

We evaluate UEMBED on text and multimodal embedding benchmarks, with results summarized in Figure 1(c). On MMEB-v2, UEMBED-9B reaches 71.8 (dense) and 71.0 (sparse), closing the gap between the two modes to within one point at every scale we test. On BEIR (9 datasets, nDCG@10), UEMBED remains competitive with strong dense and sparse baselines such as SPLADE-v3 (Lassance et al., 2024). Moreover, our analysis surfaces three practical advantages of UEMBED. First, hybrid scoring over its dense and sparse modes consistently improves text and visual-document retrieval. Second, its design retains compatibility with high-throughput serving stacks (*e.g.,* vLLM) and inverted-index search. Finally, on BrowseComp-Plus (Chen et al., 2025), the sparse mode reduces tool-call costs while maintaining comparable recall.

We summarize our contributions as follows:

- We propose UEMBED, which produces both sparse and dense retrieval representations from a single decoder-only backbone.
- UEMBED unifies different modalities in one model and can be easily extended to diverse modality settings. To our knowledge, UEMBED is the first sparse model on the MMEB-v2 leaderboard and achieves the state-of-the-art performance against the open-source models.
- Our comprehensive evaluation demonstrates that the sparse mode of UEMBED improves both search efficiency and effectiveness, while its strong performance in agentic search highlights its broad practical utility.

2 Related Works

2.1 Learned sparse retrieval

Learned sparse retrieval replaces fixed lexical statistics, such as BM25, with neural models that both

reweight observed terms and expand the representation with semantically related terms absent from the original text. SPLADE-based methods demonstrate that contextual token weighting and vocabulary expansion can make sparse retrieval competitive with strong dense baselines (Formal et al., 2021b,a). Recent work extends this line of research using LLM backbones and stronger training recipes (Xu et al., 2026). While related efforts either adapt decoder backbones toward bidirectional behavior for retrieval (Lassance et al., 2024) or jointly train dense and sparse retrieval within one model (Chen et al., 2024; Song et al., 2025), our work differs by studying sparse retrieval natively under causal attention within a unified backbone.

2.2 Multimodal Retrieval

Recently, Multimodal Large Language Models (MLLMs) (Google, 2024; Qwen Team, 2025; Singh et al., 2025) have demonstrated exceptional performance across a wide range of tasks. Consequently, universal multimodal embedding models (Jiang et al., 2024; Cui et al., 2025; Li et al., 2026) have been built upon these MLLM backbones, achieving highly competitive results on benchmarks (Meng et al., 2025; Song et al., 2026; Huang et al., 2026). However, multimodal sparse retrieval remains much less explored. Existing LSR work typically introduces auxiliary modules or an extra model to interpret non-text modalities (Lu et al., 2021; Chen et al., 2023; Nguyen et al., 2024). Such auxiliary modules restrict the sparse model to a specific modality pair and complicate its extension to new modalities. Our work addresses this gap by combining decoder-native sparse retrieval, unified dense and sparse modeling, and multimodal support within a single causal backbone.

3 Method

Figure 1 illustrates the overall framework of UEMBED. We first review preliminaries on contrastive learning and learned sparse retrieval in §3.1, then present the proposed method in §3.2, and finally describe the data curation in §3.3.

3.1 Preliminaries

InfoNCE Loss. Given a query q, a positive document d^+, and a set of in-batch negatives $\{d_j^-\}_{j=1}^K$, the InfoNCE loss encourages the model to assign higher similarity to the positive pair:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp\big(\text{sim}(q, d^+)/\tau\big)}{\sum_{d\in\{d^+\}\cup\mathcal{N}} \exp\big(\text{sim}(q, d)/\tau\big)}, \tag{1}$$

where $\mathcal{N}$ is the set of in-batch negatives, $\text{sim}(\cdot,\cdot)$ denotes a similarity function (*e.g.,* cosine similarity for dense retrieval, inner product for sparse retrieval), and τ is a temperature hyperparameter.

Learned Sparse Retrieval. SPLADE-based models (Formal et al., 2021b,a) produce a sparse representation by projecting each token's hidden state onto the vocabulary and aggregating across the sequence with a max-pooling and saturating activation:

$$w_t = \max_{i=1}^{L} \log\big(1 + \text{ReLU}(\boldsymbol{h}_i^\top \boldsymbol{e}_t)\big), \tag{2}$$

where $\boldsymbol{h}_i$ is the hidden state of the i-th input token, $\boldsymbol{e}_t$ is the embedding of vocabulary term t, L is the sequence length, and w_t is the resulting weight for term t. The sparse vector $\boldsymbol{w} \in \mathbb{R}^{|V|}$ can be served by a standard inverted index.

3.2 UEMBED Method

The key challenge in adapting sparse retrieval to decoder-only models is that unidirectional attention prevents each position from attending to future tokens, rendering max-pooling over all hidden states ineffective. We address this by introducing N special tokens appended to the input sequence and partitioning the vocabulary into N disjoint subsets.

Compressing Vocabulary. LLM tokenizers often contain redundant tokens (*e.g.,* whitespace, accents). Before partitioning, we compress the vocabulary via accent stripping, lowercasing, and whitespace collapsing using NLTK (Bird and Loper, 2004). Tokens sharing the same canonical form are merged into a single entry. For example, Hello, HELLO, and héllò are all combined into hello. During scoring, we retain the maximum weight among them. This reduces the vocabulary size from 248,320 to 184,016.

Partitioned Sparse Heads. Given an input sequence of length L, we append N=16 learnable special tokens $\langle s_1\rangle, \ldots, \langle s_N\rangle$ at the end. Each special token can attend to all preceding tokens, effectively summarizing the full input. To encourage each special token to capture a distinct semantic subspace, we employ k-means clustering to partition the vocabulary V into N disjoint subsets $V_1, \ldots, V_N$ of approximately equal size.

Each special token $\langle s_k \rangle$ is responsible for producing sparse weights over its assigned subset V_k via a subset-specific sparse head. Let H_{s_k} denote the final hidden state of $\langle s_k \rangle$. We compute the sparse weight for a vocabulary term $t \in V_k$ using a linear projection:

$$w_t^{(k)} = \log\big(1 + \mathrm{ReLU}(\boldsymbol{W}_t^{(k)\top} \boldsymbol{h}_{s_k} + b_t^{(k)})\big), \quad (3)$$

for all $t \in V_k$, where $\boldsymbol{W}_t^{(k)}$ and $b_t^{(k)}$ are the specific weight vector and bias scalar corresponding to term t in the k-th sparse head. The sparse representation is obtained by concatenating all subset vectors:

$$\boldsymbol{w} = \big[w_t^{(1)}\big]_{t \in V_1} \oplus \cdots \oplus \big[w_t^{(N)}\big]_{t \in V_N}, \quad (4)$$

Dense Representation. For dense retrieval, we use the hidden state of the EOS token preceding the special tokens as the dense embedding $\boldsymbol{d} \in \mathbb{R}^D$. When only dense retrieval is needed, the special tokens can be omitted entirely, incurring no extra forward computation.

Unified Objective. We train the model with a combined loss over both retrieval modes:

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\text{InfoNCE}}^{\text{dense}} + \lambda\, \mathcal{L}_{\text{InfoNCE}}^{\text{sparse}} \\ + \alpha_q\, \mathcal{L}_{\text{FLOPS}}^{q} + \alpha_d\, \mathcal{L}_{\text{FLOPS}}^{d}, \end{aligned} \quad (5)$$

where $\mathcal{L}_{\text{InfoNCE}}^{\text{dense}}$ and $\mathcal{L}_{\text{InfoNCE}}^{\text{sparse}}$ are computed using cosine similarity and inner product, respectively. $\mathcal{L}_{\text{FLOPS}}^{q}$ and $\mathcal{L}_{\text{FLOPS}}^{d}$ are FLOPS regularizers (Paria et al., 2020) that penalize the squared mean term weight across queries and documents respectively, encouraging sparsity. λ, α_q, and α_d are scalar coefficients balancing the four losses. The training details are provided in Appendix A.

3.3 Data Curation

We construct our training data from three publicly available sources.

Data Sources. We draw training pairs from three publicly available datasets, comprising a total of 3.94M samples. (1) **Echo-Embeddings training data** (Springer et al., 2025), which provides large-scale query–document pairs covering a wide range of domains; (2) **MLDR training data** (Chen et al., 2024), from which we select only the MLDR subset for training; and (3) **MMEB training sets** (Meng et al., 2025), which supply multimodal query–document pairs across diverse visual and cross-modal tasks.

Hard Negative Mining. As demonstrated by previous work (Formal et al., 2022), hard negatives are crucial for training learned sparse retrievers. Because the multimodal datasets lack the negatives required for effective training, we utilize Qwen3-VL-Embedding-8B (Li et al., 2026) as a teacher model to mine hard negatives for all multimodal data. Specifically, for each query, we use the teacher model to retrieve the top-k most similar non-relevant documents from the corpus, treating these as our hard negatives.

4 Experiments

4.1 Experimental Setup

Backbones and Model Sizes. We instantiate UEMBED from the Qwen3.5 family at three scales: 2B, 4B, and 9B parameters. For each scale, we train a single checkpoint that supports both dense and sparse retrieval.

Multimodal Benchmark and Baselines. We evaluate UEMBED on MMEB-V2 (Meng et al., 2025), which provides a comprehensive evaluation for multimodal embedding tasks. To comprehensively position our model within the current landscape, we compare it against a diverse selection of both open-source and proprietary baselines: (1) Qwen3-VL-Embedding (Li et al., 2026), a strong vision-language embedding baseline trained with large-scale data. (2) RzenEmbed (Jian et al., 2025), for which we adopt the competitive RzenEmbed-V2-7B variant; (3) Embed-RL (Jiang et al., 2026), which both the 2B and 4B versions are selected; (4) Ops-MM-Embed (OpenSearch-AI, 2025) and (5) UniME (Gu et al., 2025), where we select both the 2B and 7B versions for a multi-scale comprehensive comparison; (6) GME-7B (Zhang et al., 2024), a robust 7B parameter multimodal retriever; and (7) VLM2Vec (Meng et al., 2025), specifically evaluating the VLM2Vec-V2-2.0B version.

Text Benchmark and Baselines. We evaluate UEMBED on nine datasets from BEIR and adopt nDCG@10 as the evaluation metric. Details of the evaluation setup are provided in Appendix B. To ensure a fair comparison, we specifically select multimodal embedding models as our dense baselines, intentionally excluding purely text-focused models. Furthermore, we incorporate competitive sparse models to establish a comprehensive evaluation framework. Our selected baselines include: (1) GME-7B (Zhang et al., 2024), a robust 7B

Model	Image					Video					VisDoc					All
	CLS	QA	RET	GD	Avg.	CLS	QA	RET	MRET	Avg.	VDRv1	VDRv2	VR	OOD	Avg.	
# of Datasets	10	10	12	4	36	5	5	5	3	18	10	4	6	4	24	78
Qwen3-VL-Embedding-8B	74.2	81.1	80.2	92.3	80.1	78.4	71.0	58.7	56.1	67.2	87.2	69.9	88.7	73.3	82.4	77.8
Qwen3-VL-Embedding-2B	70.3	74.3	74.8	88.5	75.0	71.9	64.9	53.9	53.3	61.9	84.4	65.3	86.4	69.4	79.2	73.2
RzenEmbed-V2-7B	**70.6**	71.7	**78.5**	**92.1**	**75.9**	58.8	63.5	51.0	45.5	55.7	**89.7**	60.7	**88.7**	34.7	75.5	71.1
Embed-RL-4B	63.7	70.5	71.3	91.3	71.2	57.6	58.4	45.1	49.5	53.0	80.2	53.4	84.9	67.1	74.7	68.1
Embed-RL-2B	62.8	67.9	68.6	90.4	69.2	57.0	55.9	45.1	49.4	52.1	80.0	52.0	84.6	65.7	74.1	66.8
Ops-MM-Embed-7B	69.7	69.6	73.1	87.2	72.7	59.7	62.2	45.7	43.2	53.8	80.0	59.6	79.3	33.7	68.7	67.1
Ops-MM-Embed-2B	68.1	65.1	69.2	80.8	69.0	53.6	55.6	41.8	33.7	47.6	76.4	53.2	77.6	32.5	65.5	63.0
UniME-7B	67.1	69.2	71.9	84.8	71.2	48.6	60.7	38.2	39.3	47.5	75.7	50.5	83.7	29.1	65.7	64.1
UniME-2B	64.8	62.8	67.6	77.2	66.6	44.3	51.0	32.9	39.7	42.2	72.4	46.2	79.2	28.9	62.5	59.7
GME-7B	57.6	34.6	71.2	59.5	55.9	37.3	50.3	28.3	37.5	38.4	89.5	55.5	85.0	34.7	73.6	57.3
VLM2Vec-V2-2.0B	62.9	56.4	69.6	77.1	64.9	39.2	34.7	28.4	37.5	34.7	74.4	44.6	79.3	31.2	63.5	57.5
UEMBED-2B (dense)	59.2	67.9	68.8	85.6	67.8	56.3	52.8	45.8	41.8	50.0	82.3	59.9	85.6	67.9	77.0	66.5
UEMBED-2B (sparse)	58.9	66.1	68.5	85.1	67.0	53.7	49.5	44.2	40.3	47.7	82.3	60.6	84.5	67.3	76.7	65.5
UEMBED-4B (dense)	62.3	72.3	72.6	88.4	71.4	61.4	65.0	50.1	47.9	57.0	84.1	**61.0**	87.3	**70.4**	78.8	70.4
UEMBED-4B (sparse)	61.9	70.2	72.3	88.0	70.6	61.8	63.3	48.3	47.1	56.0	84.3	60.6	87.2	69.5	78.6	69.7
UEMBED-9B (dense)	64.7	**73.9**	73.9	90.6	73.2	**63.0**	**65.6**	**51.8**	**53.5**	**59.0**	85.5	59.1	87.9	**70.4**	**79.2**	**71.8**
UEMBED-9B (sparse)	64.3	72.3	73.7	89.5	72.5	60.9	63.7	50.9	52.4	57.5	86.2	57.9	87.6	69.7	79.1	71.0

Table 1: Performance comparison on various benchmarks.

multimodal embedding model that demonstrates strong text retrieval performance on BEIR; (2) Qwen3-VL-Embedding (Li et al., 2026), a powerful vision-language embedding baseline trained on large-scale data that also exhibits competitive text-only retrieval capabilities; (3) SPLADE-v3 (Lassance et al., 2024), representing the latest iteration of the classical sparse retrieval method; and (4) Echo-Mistral-SPLADE (Doshi et al., 2024), which utilizes a powerful LLM backbone to generate highly effective sparse representations.

4.2 Multimodal Results

UEMBED is competitive with the strongest open multimodal embedders at every scale it operates. As shown in Table 1, UEMBED-9B (dense) reaches an aggregate score of 71.8. While this is surpassed by Qwen3-VL-Embedding-8B, it is crucial to note that Qwen's models benefit from multi-stage training with vast, proprietary datasets, making a direct comparison challenging. When compared with other models that have publicly available checkpoints and are trained on open datasets, UEMBED-9B (dense) leads models trained on publicly available data, outperforming models like RzenEmbed-V2-7B (71.1) and Ops-MM-Embed-7B (67.1). At smaller scales, the picture is consistent: UEMBED-4B (dense) with a score of 70.4 surpasses every other model in the 4B-parameter class, such as Embed-RL-4B (68.1). Furthermore, our most compact model, UEMBED-2B (dense) at 66.5, is highly competitive with, and even outperforms some, 7B-scale open baselines like UniME-7B (64.1). To better understand the gap between UEMBED and Qwen3-VL-Embedding, we also conduct an analysis in Appendix C.

Sparse retrieval remains competitive with dense retrieval in the multimodal regime. Our results demonstrate that sparse embeddings are a viable and powerful alternative to dense ones in the multimodal domain. As shown in the table, the performance of our sparse models is remarkably close to their dense counterparts across all scales, with the overall performance gap being at most 1.0 point (*e.g.,* 71.8 vs. 71.0 for the 9B model). To our knowledge, these are the first reported results for sparse multimodal embeddings on this benchmark, and they significantly outperform established dense models. For instance, UEMBED-4B (sparse) at 69.7 surpasses the dense Ops-MM-Embed-7B (67.1). A closer look reveals that sparse models are particularly effective on Visually-rich Document (VisDoc) tasks, where the performance drop is minimal (*e.g.,* 79.2 vs. 79.1 for the 9B model). This suggests that the inherent structure of sparse embeddings may be well-suited for document-based tasks, while the performance gap is slightly more pronounced in the Video category.

Model	ArguAna	FiQA 2018	NFCorpus	NQ	Quora	SCIDOCS	SciFact	COVID	Touche 2020	Avg
Dense Models										
GME-7B	**64.6**	**57.1**	38.4	**67.7**	88.1	**27.4**	62.3	52.6	23.3	53.5
Qwen3-VL-Embedding-2B	43.2	41.8	37.2	58.1	86.6	21.9	75.1	**89.6**	**30.8**	53.8
Qwen3-VL-Embedding-8B	45.8	47.8	39.7	64.5	86.5	24.9	**79.4**	84.1	26.9	55.5
UEMBED-2B	58.1	47.5	37.8	55.5	89.3	20.3	75.3	80.2	18.2	53.6
UEMBED-4B	60.0	53.4	**40.5**	60.9	89.2	22.5	77.0	82.7	18.1	56.0
UEMBED-9B	62.5	54.3	39.8	61.9	**89.9**	23.4	77.8	77.9	19.2	**56.3**
Sparse Models										
SPLADE-v3	50.9	37.4	35.7	58.6	81.4	15.8	71.0	74.8	**29.3**	50.5
Echo-Mistral-SPLADE	56.2	**57.7**	**42.3**	56.0	86.7	**25.6**	**77.2**	76.8	18.0	**55.2**
UEMBED-2B	55.7	45.1	35.9	53.4	88.2	18.0	70.8	**84.8**	18.7	52.3
UEMBED-4B	**58.8**	49.4	38.4	**59.9**	88.6	20.8	74.4	75.4	16.7	53.6
UEMBED-9B	58.4	51.2	38.1	**59.9**	**88.7**	20.6	74.5	82.1	23.4	**55.2**

Table 2: nDCG@10 on BEIR datasets. **Bold** marks the best score in each column within each retrieval mode.

4.3 Text Results

In the dense retrieval setting, as shown in Table 2, UEMBED demonstrates highly competitive performance across the evaluated benchmarks. Our UEMBED-9B model achieves the highest average nDCG@10 score of **56.3** among the compared models, with UEMBED-4B closely following at 56.0. This places our models ahead of strong, recent baselines like Qwen3-VL-Embedding-8B and GME-7B. Furthermore, UEMBED shows particular strength on specific datasets, such as achieving 89.9 on Quora (UEMBED-9B) and 40.5 on NFCorpus (UEMBED-4B). In the sparse retrieval setting, UEMBED-9B reaches an average of **55.2**, effectively matching the strong specialist model Echo-Mistral-SPLADE (55.2). Crucially, UEMBED achieves this parity while simultaneously supporting dense retrieval and multimodal inputs within a single backbone. These results position UEMBED as a unified model that sacrifices little text performance while gaining robust multimodal and dense capabilities.

5 Analysis

To better understand UEMBED, this section analyzes the model from the following perspectives: (1) **Component Analysis:** controlled ablations to validate the contribution of individual design choices (§5.1); (2) **Cross-modal Robustness:** case studies illustrating the model's performance across diverse modalities (§5.2); and (3) **Practical Advantages:** the utility of sparse retrieval in hybrid scoring, serving efficiency, and agentic search.

5.1 Component Analysis

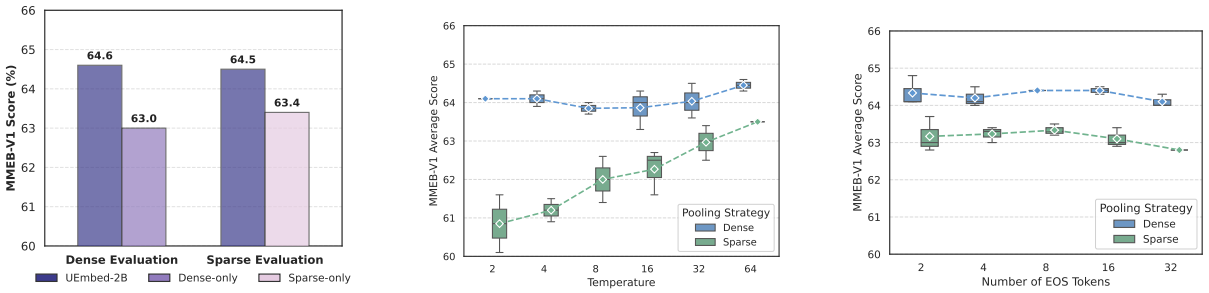
Unless otherwise noted, ablations in this section use the 2B backbone trained on a random 500k-instance subset from the full mixture, and are evaluated on the MMEB-v1 (Jiang et al., 2024).

	CLS	QA	RET	GRD	Avg.
SPLADE	55.8	57.1	62.9	80.9	61.3
UEMBED $_{\text{dense}}$	55.5	65.3	65.1	83.3	64.5
UEMBED $_{\text{sparse}}$	54.7	63.1	64.8	81.8	63.4

Table 3: Comparison with the SPLADE baseline on the image subsets of MMEB-v1. Both the dense and sparse modes of UEMBED-2B outperform SPLADE.

Baseline Comparison. To determine if the gains in §4.2 stem from the UEMBED design, we compare UEMBED-2B against a SPLADE-based baseline. This baseline shares the same Qwen3.5 backbone, training data, and FLOPS regularization, but adopts bidirectional attention with a standard SPLADE max-pooling head (Equation 2). Table 3 demonstrates that UEMBED-2B surpasses this bidirectional baseline in *both* modes: $+3.2$ points for dense ($61.3 \rightarrow 64.5$) and $+2.1$ for sparse ($61.3 \rightarrow 63.4$) on the average. The margins are most pronounced on IMG-QA ($+8.2$ dense, $+6.0$ sparse), suggesting the standard SPLADE recipe underutilizes the autoregressive backbone's inherent QA capabilities. Thus, our unified causal formulation is intrinsically beneficial to embedding quality, beyond merely offering serving convenience.

Robustness to Joint Training. A common pitfall in multi-task learning is negative transfer between objectives. To verify if our joint objective (Equa-



(a) Joint vs. single-mode training. (b) Sparse temperature τ_s. (c) Number of special tokens N.

Figure 2: Ablation studies of UEMBED: (a) joint training matches single-mode specialists; (b) sweeping the sparse softmax temperature τ_s; and (c) sweeping the number of appended special tokens N.

tion 5) compromises either retrieval mode, we train dense-only (λ=0) and sparse-only baselines under identical conditions. As shown in Figure 2a, the jointly trained UEMBED model matches the performance of both single-mode specialists. This confirms that the dual-mode capability incurs zero performance penalty during training.

Partition	CLS	QA	RET	GRD	Avg.
Random	55.0	62.3	64.2	81.0	63.0
Max-distance	55.1	62.4	64.9	79.8	63.2
Semantic (ours)	54.7	63.1	64.8	81.8	63.4

Table 4: Sparse performance under different vocabulary partitioning strategies on MMEB-v1.

Vocabulary Partitioning. The sparse head partitions the vocabulary V into N disjoint subsets. We compare three balanced partitioning strategies: (1) **Random** uniform assignment; (2) **Max-distance**, which pairs maximally distant k-means clusters to enforce semantic diversity within subsets; and (3) **Ours (Semantic clustering)**, which assigns individual k-means clusters to distinct subsets. Semantic clustering yields the best performance, while max-distance performs best. This suggests that grouping semantically related tokens allows each special token to act as a "soft topic specialist," optimizing its limited representational capacity rather than struggling to model unrelated concepts simultaneously.

Hyperparameter Sensitivity. We sweep two hyperparameters specific to the sparse head: *Sparse temperature (τ_s).* Unlike the dense head's $[-1, 1]$ cosine similarities, sparse inner products have a much larger dynamic range. A shared temperature would therefore over-sharpen the sparse softmax. Figure 2b demonstrates that decoupling the temperatures and using a larger τ_s (set to 32) improves sparse performance without degrading dense retrieval. *Number of special tokens (N).* Sweeping $N \in \{2, 4, 8, 16, 32\}$ (Figure 2c) reveals stable performance up to N=16, with a slight drop at N=32. We attribute this degradation to overly small vocabulary subsets and inflated contrastive sequence lengths. We default to N=16 to optimally balance expressivity and training efficiency.

5.2 Case Study

Building upon the examples in Figure 1(b), we present additional case studies across various modalities. As illustrated in Figure 3, our model demonstrates a robust capacity to identify essential components within the input and activate semantically relevant concepts. For instance, it can deduce an image's location as Singapore based on its skyline and accurately identify a rocket as belonging to SpaceX. However, we also observe two primary limitations: (1) *Generation of Anomalous Tokens*: The model occasionally produces non-standard tokens (*e.g.,* "_alt"). Unlike traditional masked language models such as BERT, which operate within a restricted and highly structured vocabulary, the exceptionally large vocabulary size of modern LLMs increases the probability of generating such edge-case tokens. This suggests that further vocabulary compression or targeted pruning may be necessary to ensure representational stability. (2) *Limited Multilingual Support*: The model predominantly activates English and Chinese tokens while neglecting other languages. This bias primarily stems from our training corpus, which is heavily skewed toward English and Chinese, as well as the inherent cross-lingual limitations of the underlying foundation model.

Model	Accuracy (%) ↑	Recall (%) ↑	Avg. Search ↓	Calibration Error (%)
BM25	36.87	43.02	17.91	–
Qwen3-Embedding-8B	44.46	62.32	30.37	–
Qwen3-VL-Embedding-2B	26.87	38.20	34.99	12.13
Qwen3-VL-Embedding-8B	31.45	44.42	34.19	12.35
UEMBED-2B (dense)	44.10	53.47	39.19	16.69
UEMBED-2B (sparse)	44.05	57.04	32.67	8.54
UEMBED-4B (dense)	51.57	61.03	38.47	14.34
UEMBED-4B (sparse)	45.54	60.10	31.85	8.79
UEMBED-9B (dense)	49.76	64.72	33.68	7.06
UEMBED-9B (sparse)	49.76	64.33	31.05	8.16

Table 5: Results on BrowseComp-Plus. "Avg. Search" denotes the average number of search rounds per question.

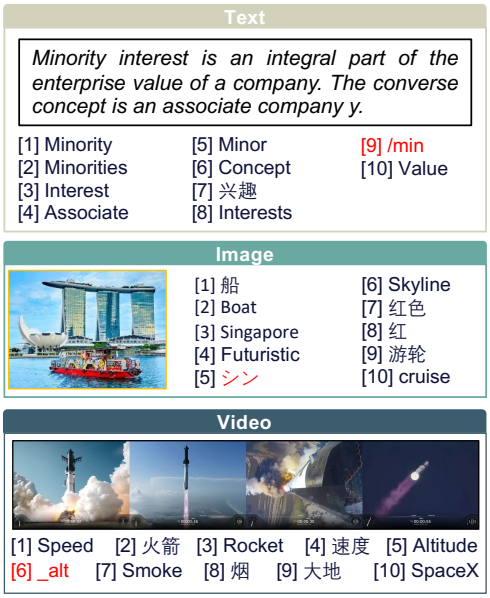


Figure 3: Case study of UEMBED on different modalities. We visualize the top-10 activated sparse tokens.

5.3 Practical Advantages

In this section, we examine three aspects of UEMBED: (1) the *effectiveness* gain from combining its sparse and dense modes via hybrid scoring; (2) the deployment-side *efficiency* enabled by autoregressive serving and native inverted-index compatibility ; and (3) its utility as a retriever in downstream *agentic search* workflows.

Model	TXT	IMG	VID	VDR
UEMBED (dense)	53.6	67.8	50.0	77.0
UEMBED (sparse)	52.3	67.0	47.7	76.7
UEMBED (hybrid)	53.9	67.9	50.0	77.5

Table 6: Modality-wise performance of UEMBED-2B under dense, sparse, and hybrid mode.

Effectiveness. Because UEMBED produces sparse and dense representations in a single forward pass, the two modes can be combined at near-zero additional cost. We construct a hybrid score by linearly interpolating the dense and sparse similarities; the detailed formulation and per-modality weights are provided in Appendix B.3. Table 6 shows hybrid scoring boosts *Text* (+0.3) and *VisDoc* (+0.5), where exact lexical matching complements dense semantics. In contrast, natural images and video frames carry little surface-form lexical information, leaving sparse retrieval with limited additional signal to contribute. Although sparse retrieval provides potential gains on lexically-dependent tasks, the aggregate improvement appears modest due to dilution across the diverse benchmarks.

Efficiency. UEMBED offers two deployment-side advantages. First, because the model is purely autoregressive, embedding generation is directly compatible with high-throughput serving stacks such as vLLM (Kwon et al., 2023). Second, the sparse representations plug into standard inverted indices, enabling scalable lexical retrieval in production environments. We defer a detailed comparison of generation throughput and inverted-index search efficiency to Appendix B.3.

Application: Agentic Search. As noted by Meng et al. (2026), LLM agents frequently issue short, keyword-dense queries, a regime where sparse retrieval typically outperforms dense alternatives. To further explore this problem, we test UEMBED on BrowseComp-Plus (Chen et al., 2025), a benchmark where an LLM uses retrieval as a tool during iterative reasoning. We utilize DeepResearch-30A3B (Team, 2025) as the backbone reasoning engine. As shown in Table 5, the sparse mode of UEMBED consistently requires fewer tool-call search rounds than its dense counterpart. Recall is higher at the 2B scale and com-

parable at 4B and 9B. These results validate sparse retrieval as a practical choice for cost-sensitive, iterative reasoning loops where keyword-dense queries dominate (Zhao et al., 2026).

6 Conclusion

In this work, we present UEMBED, a decoder-only multimodal embedding model that natively unifies dense and sparse retrieval within a single causal forward pass.By appending N learnable special tokens and partitioning the vocabulary into disjoint subsets, UEMBED successfully overcomes the single-token representational bottleneck inherent to causal attention, systematically eliminating the historical reliance on legacy bidirectional encoders. Through rigorous component analysis, we validate our core algorithmic designs. Our comprehensive evaluations demonstrate that UEMBED establishes a new state-of-the-art for open-source multimodal retrieval in both dense and sparse mode. Crucially, its architectural elegance ensures out-of-the-box compatibility with modern high-throughput serving frameworks (*e.g.,* vLLM) and traditional inverted indices. This efficiency, coupled with tangible cost reductions in agentic search pipelines, positions UEMBED as a highly scalable, paradigm-shifting approach to modern information retrieval. Overall, it transforms sparse retrieval from a legacy, computationally expensive standalone module into a native, highly efficient byproduct of MLLMs.

Limitations

While UEMBED establishes a strong foundation for unified multimodal retrieval, we identify several areas for future improvement. (1) *Language and Cultural Bias:* The training corpus utilized in this work is predominantly skewed toward English and Chinese. Consequently, the sparse head activations exhibit limited cross-lingual generalization. (2) *Vocabulary Stability and Artifacts:* Unlike traditional masked language models with constrained vocabularies, operating over the expansive vocabulary of modern LLMs leads to anomalous token activations (*e.g.,* non-standard subwords or artifacts like "_alt"). (3) *Modality-Specific Performance Gaps:* While the sparse and dense representations maintain close performance parity on text and static visual documents, we observe a slightly more pronounced gap in the video domain. This suggests that pooling spatiotemporal data into a sparse vector might encounter capacity bottlenecks.

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A Implementation Details

A.1 Training Data

We curate a multi-source training mixture that covers text, image, video, and visually-rich document retrieval. The composition ratio for each data source is shown in Figure 4, and per-dataset retrieval instructions are listed in Table 7 and Table 8.

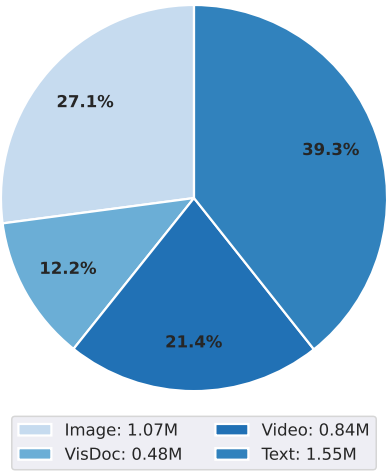


Figure 4: Distribution of training data sources.

Text Training Data. To enable text embedding capabilities, we include training pairs from Echo-Embeddings (Springer et al., 2025) and M3-Embedding (Chen et al., 2024). The text subset comprises NLI, DuReader, ELI5, FEVER, HotpotQA, MIRACL, MrTyDi, MSMARCO Passage, MSMARCO Document, Natural Questions, Quora Duplicates, SQuAD, T2Ranking, TriviaQA, and MLDR. For standard text tasks we set a maximum sequence length of 1,500 tokens; for long-document retrieval (MLDR) we extend this to 1,800 tokens.

Multimodal Training Data. For video retrieval, we utilize VideoCaption300k and VideoQA240k from LLaVA-Hound (Zhang et al., 2025). We sample 8 frames per video and allocate up to 200 tokens per frame. For visual document retrieval, we incorporate ColPali train (118k), VisRAG-Synthetic (239k), and VisRAG-IID (123k) from ViDoRe (Faysse et al., 2025) and VisRAG (Yu et al., 2025). For image-text retrieval, we include the MMEB training datasets (Jiang et al., 2024), covering CIRR, NIGHTS, MSCOCO, VisualNews, N24News, ImageNet-1K, SUN397, VOC2007, HatefulMemes, A-OKVQA, OK-VQA, Visual7W, ChartQA, DocVQA, InfographicsVQA, VisDial, and WebQA. For video tasks, the maximum sequence length is 1,800 tokens with 8 frames sampled per video and up to 200 tokens per frame. For all other multimodal tasks, the maximum sequence length is 1,500 tokens, with up to 1,000 tokens allocated to the image.

A.2 Training Details

We initialize from a pretrained multimodal backbone and add N=16 special tokens to the tokenizer and embedding matrix. Training uses LoRA (Hu et al., 2022) applied to the attention and MLP projections (`q_proj`, `k_proj`, `v_proj`, `up_proj`, `down_proj`, `gate_proj`), with the visual encoder frozen. We train in `bf16` mixed precision with DeepSpeed ZeRO and gradient checkpointing enabled.

Hyperparameters. We train the UEMBED models based on Qwen3.5. We use a cosine learning rate schedule with a peak learning rate of 3×10^{-5} and a warmup ratio of 0.1. The dense retrieval temperature is set to τ=0.03, and the sparse retrieval temperature is τ_s=32. For the FLOPs regularizer, we set $\alpha_q = \alpha_d = 1\times10^{-4}$ with a linear ramp-up over the first 200 steps. The sparse loss factor λ=1.0 balances the sparse and dense InfoNCE losses equally. We train for 1 epoch and fix the random seed to 42. Training is conducted on 16×A100 GPUs for UEMBED-2B and UEMBED-4B with a per-device batch size of 16, and on 32×A100 GPUs for UEMBED-9B with a per-device batch size of 8, ensuring a consistent global batch size of 256 across all models.

A.3 Training Analysis

Since sparse scores are computed via inner products, sparse training can be unstable due to the large dynamic range of unnormalized similarities. We find that incorporating text training data significantly stabilizes and accelerates multimodal sparse model training. Following GVE (Guo et al., 2025), we monitor three key metrics during training: loss, max negative gap, and mean negative gap. The max negative gap is defined as the largest score difference between a positive and the negative within a batch, while the mean negative gap is the average score difference across all positive-negative pairs in a batch. As shown in Figure 5, when trained with text data, the model learns effective sparse representations rapidly (<100 steps). In contrast, training exclusively on multimodal data (*e.g.,* im-

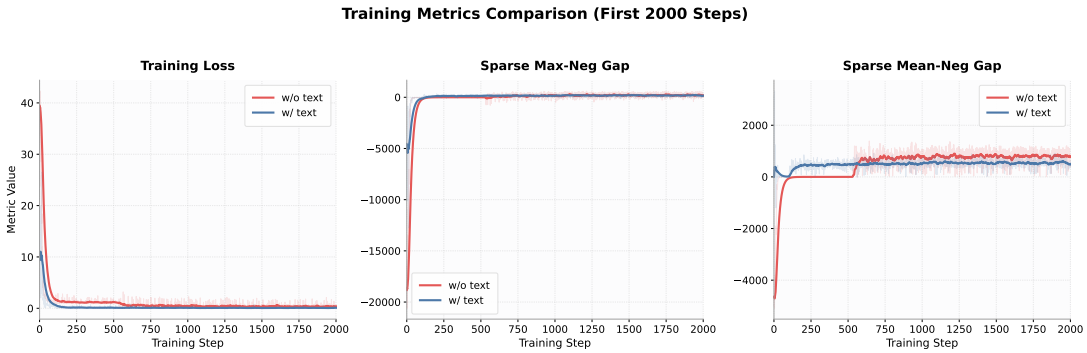


Figure 5: Training dynamics comparison. We plot the loss, max negative gap, and mean negative gap during training.

Dataset	Instruction
AllNLI	Given a premise, retrieve a hypothesis that is entailed by the premise. Retrieve semantically similar text.
DuReader	Given a Chinese search query, retrieve web passages that answer the question.
ELI5	Provided a user question, retrieve the highest voted answers on Reddit ELI5 forum.
FEVER	Given a claim, retrieve documents that support or refute the claim.
HotpotQA	Given a multi-hop question, retrieve documents that can help answer the question.
MIRACL	Given a question, retrieve Wikipedia passages that answer the question.
MrTyDi	Given a question, retrieve Wikipedia passages that answer the question.
MSMARCO-Doc	Given a web search query, retrieve relevant documents that answer the query.
MSMARCO-Psg	Given a web search query, retrieve relevant passages that answer the query.
NQ	Given a question, retrieve Wikipedia passages that answer the question.
Quora	Given a question, retrieve questions that are semantically equivalent to the input question.
SQuAD	Retrieve Wikipedia passages that answer the question.
T2Ranking	Given a Chinese search query, retrieve web passages that answer the question.
TriviaQA	Retrieve Wikipedia passages that answer the question.
MLDR	Retrieve Wikipedia passages that answer the question.

Table 7: Retrieval instructions for text training datasets.

age, video) requires nearly 500 steps before the model can distinguish positives from negatives.

B Experiment Details

B.1 Evaluated Models

We provide the details of the baselines compared against UEMBED on the multimodal benchmark MMEB-V2 (Table 1) and the text benchmark BEIR (Table 2) in Table 9.

B.2 Evaluation Settings

Metrics. For MMEB-v2 (Meng et al., 2025), we use its evaluation metrics for different datasets. For BEIR, we use nDCG@10 as the evaluation metric.

Scores Calculation. Unless otherwise noted, dense scores are cosine similarities of the last-content-token hidden state and sparse scores are inner products over the partitioned vocabulary representation defined in §3.2.

B.3 Details of Ablation Study

Effectiveness. We further detail the hybrid scoring used in §5.3. Given a query q and a candidate d, the hybrid score linearly combines the dense cosine similarity $s_{\text{dense}}(q, d)$ and the sparse inner product $s_{\text{sparse}}(q, d)$ as

$$s_{\text{hybrid}}(q, d) = \alpha\, s_{\text{dense}}(q, d) + \beta\, s_{\text{sparse}}(q, d). \quad (6)$$

Because s_{dense} and s_{sparse} lie on very different scales, dense cosine similarities are bounded in $[-1, 1]$ while sparse inner products can take much larger magnitudes. We fix $\alpha = 1.0$ and tune β per modality on a held-out split. We use $\beta = 5\text{e}{-8}$ for Text, $\beta = 5\text{e}{-4}$ for Image, $\beta = 1\text{e}{-4}$ for Video, and $\beta = 7\text{e}{-4}$ for VisDoc. The optimum for plain text is several orders of magnitude smaller because raw sparse inner products grow rapidly with the number of activated tokens, whereas the heavier multimodal inputs yield much sparser activation patterns and therefore admit larger β values.

Efficiency. Beyond generation throughput, the sparse representations of UEMBED integrate natively with inverted indices, opening the door to large-scale lexical retrieval. To quantify this, we benchmark the offline retrieval mode of BrowseComp-Plus, comparing a Faiss-based dense index against a Lucene-backed sparse inverted index. Figure 6 illustrates the latency–accuracy trade-off of our sparse mode: by capping the maximum number of activated tokens per query (N), practitioners can flexibly trade search latency for retrieval quality (NDCG@5) to match deployment constraints. While dense search yields slightly higher absolute performance at this corpus size, the comparative advantage of inverted indices is highly scale-dependent: dense retrieval remains competitive on smaller collections, whereas the throughput advantages of sparse retrieval materialize once the corpus grows to web-scale.

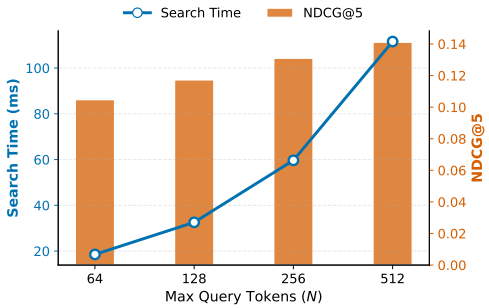


Figure 6: Search efficiency–effectiveness trade-off between dense retrieval and sparse inverted-index retrieval.

B.4 Extra Experiment Results

To provide a more reliable test of our hyperparameter settings, we report the MMEB-v2 results of models trained with random seed 42. These trends are broadly consistent with the main-paper settings: (1) joint dense-sparse training preserves dense performance while producing a competitive sparse branch Table 10; (2) using too many special tokens hurts sparse retrieval Table 11; (3) a higher sparse-training temperature is beneficial Table 12.

Dataset	Instruction
CIRR	Given an image, find a similar everyday image with the described changes.
NIGHTS	Find a day-to-day image that looks similar to the provided image.
MSCOCO	Select the portion of the image that isolates the object.
MSCOCO (image-to-text)	Find an image caption describing the given everyday image.
VisualNews (image-to-caption)	Find a caption for the news in the given photo.
MSCOCO (text-to-image)	Find me an everyday image that matches the given caption.
VisualNews (text-to-image)	Retrieve an image of this news caption.
N24News	Classify the domain of the given news image.
ImageNet_1K	Classify the given image.
SUN397	Identify the scene shown in the image.
VOC2007	Identify the object shown in the image.
HatefulMemes	Determine whether the given image constitutes hateful speech or not.
A-OKVQA	Answer the question about the given image.
OK-VQA	Answer the question about the given image.
Visual7W	Answer the question about the given image.
ChartQA	Answer the question about the given chart image.
DocVQA	Answer the question about the given document image.
InfographicsVQA	Answer the question about the given infographic image.
VisDial	Retrieve an image based on the given dialogue.
WebQA	Find a Wikipedia image that answers this question.
VISRAG-IID	Find a Wikipedia image that answers this question.
ViDoRe	Find a Wikipedia image that answers this question.
VisRAG-Synthetic	Find a document image that answers this question.
VideoCaption300k (text-to-video)	Find a video that matches the given caption.
VideoCaption300k (video-to-text)	Find a caption that describes the given video.
VideoQA240k	Answer the question about the given video.

Table 8: Retrieval instructions for multimodal training datasets.

Model	Release	Retrieval	Backbone	#Params
Baselines				
Qwen3-VL-Embedding-8B	2026-01	Dense	Qwen3-VL-8B	8B
Qwen3-VL-Embedding-2B	2026-01	Dense	Qwen3-VL-2B	2B
RzenEmbed-V2-7B	2025-10	Dense	Qwen2-VL-7B	7B
Embed-RL-4B	2026-02	Dense	Qwen3-VL-4B	4B
Embed-RL-2B	2026-02	Dense	Qwen3-VL-2B	2B
Ops-MM-Embed-7B	2025-07	Dense	Qwen2-VL-7B	7B
Ops-MM-Embed-2B	2025-07	Dense	Qwen2-VL-2B	2B
UniME-7B	2025-10	Dense	Qwen2-VL-7B	7B
UniME-2B	2025-10	Dense	Qwen2-VL-2B	2B
GME-7B	2024-12	Dense	Qwen2-VL-7B	7B
VLM2Vec-V2-2.0B	2025-05	Dense	Qwen2-VL-2B	2B
SPLADE-v3	2024-03	Sparse	BERT	110M
Echo-Mistral-SPLADE	2024-08	Sparse	Mistral-7B	7B
Our model				
UEMBED-2B	2026-08	Dense & Sparse	Qwen3.5-2B	2B
UEMBED-4B	2026-08	Dense & Sparse	Qwen3.5-4B	4B
UEMBED-9B	2026-08	Dense & Sparse	Qwen3.5-9B	9B

Table 9: Details of the baselines compared against UEMBED. Multimodal models are evaluated on MMEB-V2; text-only models are evaluated on the nine BEIR datasets reported in Table 2. UEMBED produces both dense and sparse embeddings from a single checkpoint.

Mode	Image					Video					VisDoc					All
	CLS	QA	RET	GD	Avg.	CLS	QA	RET	MRET	Avg.	VDRv1	VDRv2	VR	OOD	Avg.	
# of Datasets	10	10	12	4	36	5	5	5	3	18	10	4	6	4	24	78
Dense-only	55.3	65.4	65.3	83.5	64.6	56.3	52.2	44.8	48.3	50.6	81.6	61.7	82.9	66.8	76.1	64.9
Sparse-only	54.1	63.6	64.2	79.8	63.0	54.1	52.6	43.6	45.6	49.3	79.9	58.5	82.5	64.9	74.5	63.4
UEmbed-2B (dense)	55.5	65.3	65.1	83.3	64.5	58.3	46.9	45.6	47.5	49.8	81.6	59.5	83.1	66.7	75.8	64.6
UEmbed-2B (sparse)	54.7	63.1	64.8	81.8	63.4	56.0	46.3	44.4	45.7	48.4	79.9	59.3	82.0	65.9	74.7	63.4

Table 10: MMEB-v2 performance comparison across representation modes.

N	Image					Video					VisDoc					All
	CLS	QA	RET	GD	Avg.	CLS	QA	RET	MRET	Avg.	VDRv1	VDRv2	VR	OOD	Avg.	
# of Datasets	10	10	12	4	36	5	5	5	3	18	10	4	6	4	24	78
2	56.0	62.9	65.0	81.2	63.7	56.6	48.2	45.3	46.6	49.4	80.1	59.4	82.1	65.8	74.8	63.8
4	55.0	62.7	64.9	80.4	63.3	57.0	47.3	45.7	44.5	49.1	80.2	60.3	82.0	65.7	74.9	63.6
8	54.9	62.2	65.2	81.0	63.2	55.1	48.9	44.8	46.1	49.0	79.6	58.8	81.6	65.8	74.3	63.4
16	54.7	63.1	64.8	81.8	63.4	56.0	46.3	44.4	45.7	48.4	79.9	59.3	82.0	65.9	74.7	63.4
32	54.6	62.8	65.0	80.3	63.2	54.4	46.2	41.9	45.7	47.2	75.6	22.3	73.4	65.7	64.5	59.9

Table 11: MMEB-v2 performance under different numbers of special tokens (N).

Temperature	Image					Video					VisDoc					All
	CLS	QA	RET	GD	Avg.	CLS	QA	RET	MRET	Avg.	VDRv1	VDRv2	VR	OOD	Avg.	
# of Datasets	10	10	12	4	36	5	5	5	3	18	10	4	6	4	24	78
2	53.7	61.7	62.8	77.4	61.6	50.7	48.6	41.7	44.3	46.6	79.5	57.5	80.0	64.4	73.4	61.8
4	52.9	60.4	63.2	78.0	61.2	50.4	49.0	42.4	44.1	46.7	79.1	56.5	80.3	64.7	73.2	61.6
8	55.0	60.8	64.5	80.2	62.6	54.9	50.6	43.2	43.6	48.6	80.3	58.1	81.6	65.3	74.4	63.0
16	54.0	62.2	64.3	80.8	62.7	55.7	53.0	43.0	44.8	49.6	80.1	57.7	82.2	65.4	74.4	63.3
32	54.7	63.1	64.8	81.8	63.4	56.0	46.3	44.4	45.7	48.4	79.9	59.3	82.0	65.9	74.7	63.4
64	54.8	63.4	64.7	81.8	63.5	54.6	50.5	44.5	49.3	49.8	80.4	58.9	82.1	64.9	74.7	63.8

Table 12: MMEB-v2 performance under different temperature values.

C Case Study

C.1 Qualitative Analysis

To better understand the gap between UEMBED and Qwen3-VL-Embedding, we recomputed per-query Hit@1 on MMEB-v2 and inspected cases where Qwen3-VL-Embedding succeeds but UEmbed fails. The errors are modality-dependent: (1) Image. The gap mainly appears in fine-grained recognition (ImageNet-A, SUN397, N24News) and question-grounded VQA (GQA, ScienceQA), where UEmbed sometimes confuses visually similar categories or follows misleading lexical cues. (2) Video. Qwen3-VL-Embedding is stronger on action recognition, moment localization, and long-video reasoning (HMDB51, UCF101, EgoSchema, NExTQA, QVHighlight), likely due to its dedicated temporal/motion supervision, while our current training contains little video-specific data. (3) VisDoc. The gap is smaller and mostly appears in multilingual ViDoRe and table/numerical-reasoning documents. Qwen3-VL-Embedding likely benefits from its rank-KL objective, which distills reranker scores into the embedding model.

C.2 Case Study

We present additional case studies of UEMBED across different tasks and modalities. For each example, we visualize the top-10 activated sparse tokens. In the figures, blue tokens denote query activations, green tokens denote corpus activations, and red tokens indicate co-activated tokens shared between the query and corpus. We find that the sparse representations perform well on QA, Retrieval, and Grounding tasks, where the activated tokens are highly relevant to the query intent. However, for Classification, the model tends to over-associate with tangential concepts and struggles with instruction following, leading to less precise sparse activations.

D Compliance and Reproducibility Statement

This study adheres to transparent and ethical research practices. All datasets and models were used in strict accordance with their respective licenses. Specifically, the MMEB-v2 benchmark and the Qwen-3.5 model are governed by the Apache-2.0 License. The BEIR benchmark is used under the terms of the Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0) license. Our research methodology and the dissemination of our findings fully comply with the stipulations of these agreements.

Furthermore, in the preparation of this manuscript, we utilized large language models (LLMs) as auxiliary tools strictly for grammar checking and language polishing. All AI-generated suggestions were critically reviewed and edited by the authors, who assume full responsibility for the intellectual content and final version of this paper.

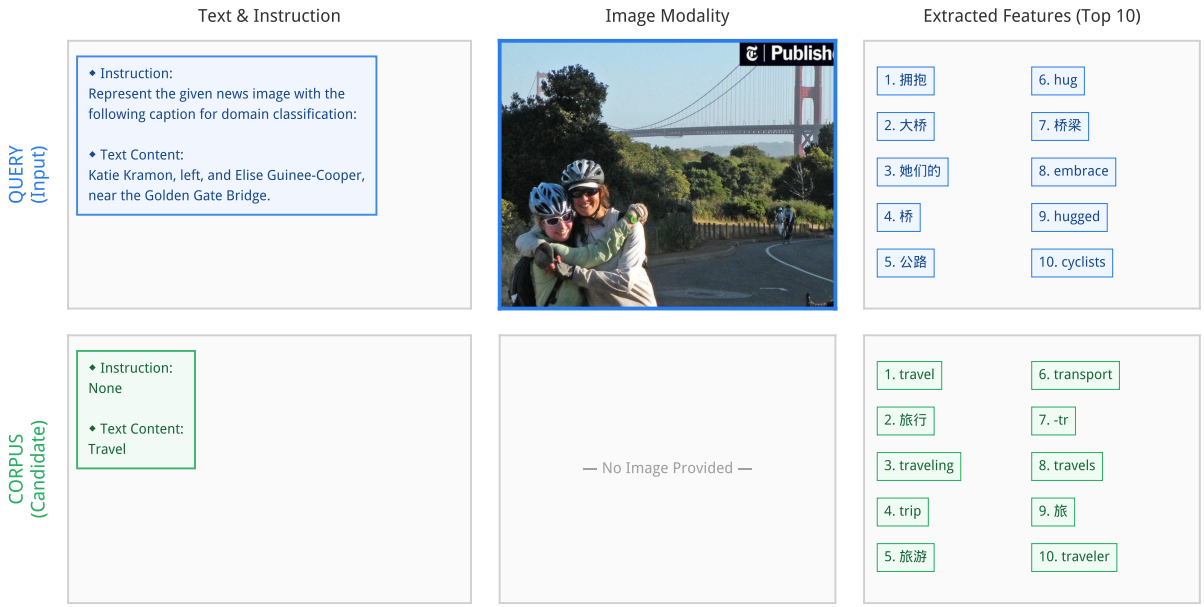


Figure 7: Case study of UEMBED on classification (N24News). We visualize the top-10 activated sparse tokens.

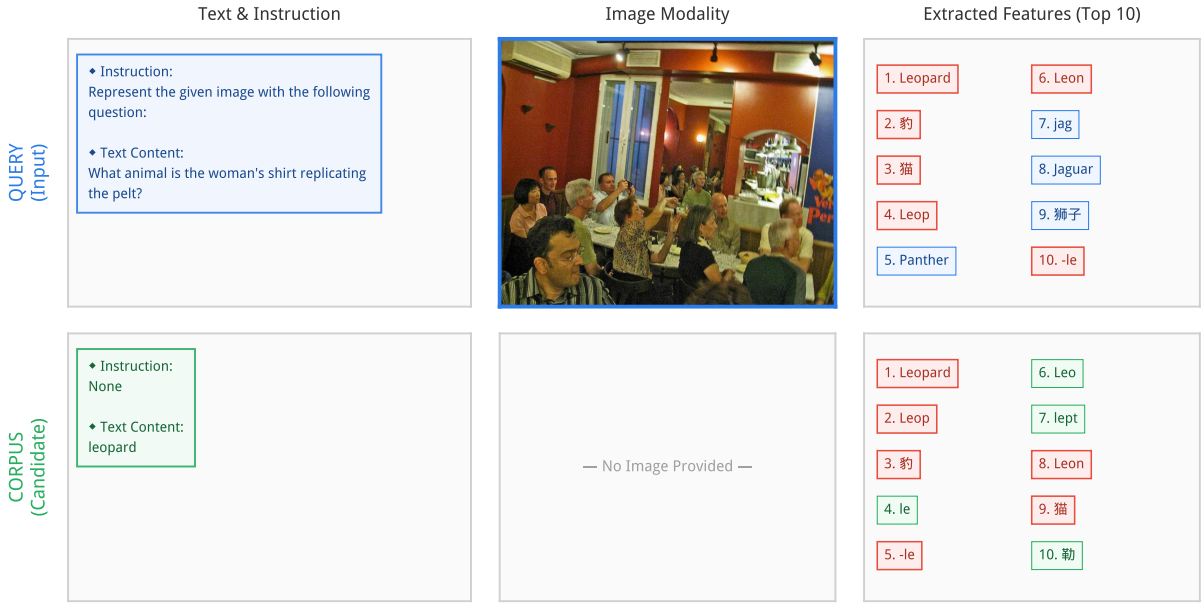


Figure 8: Case study of UEMBED on question answering (OK-VQA). We visualize the top-10 activated sparse tokens.

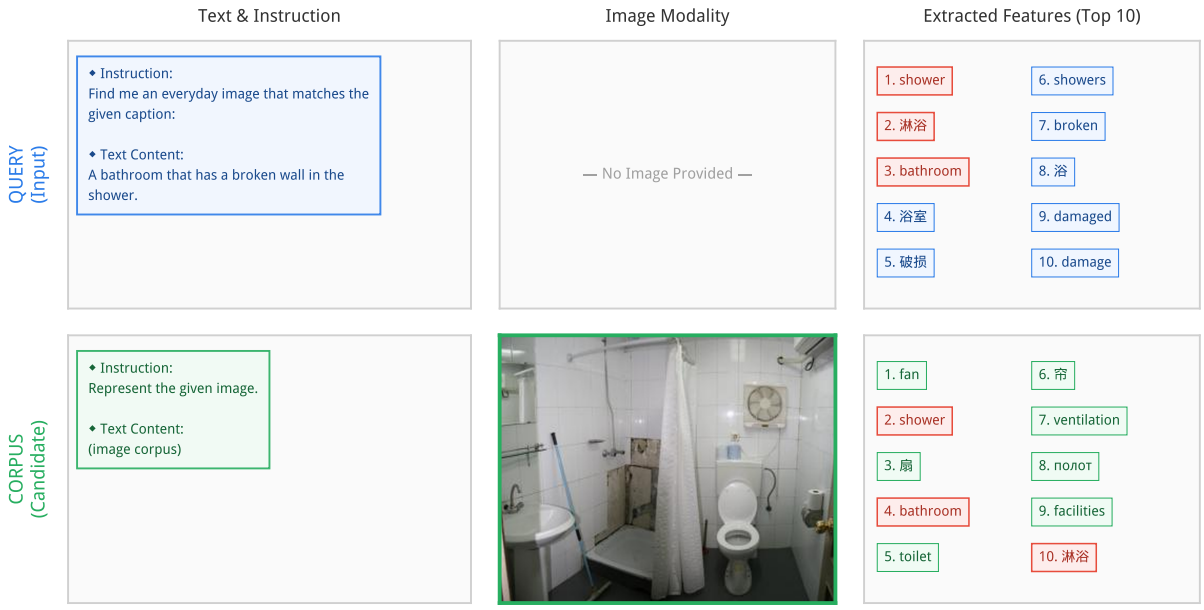


Figure 9: Case study of UEMBED on text-to-image retrieval (MSCOCO). We visualize the top-10 activated sparse tokens.

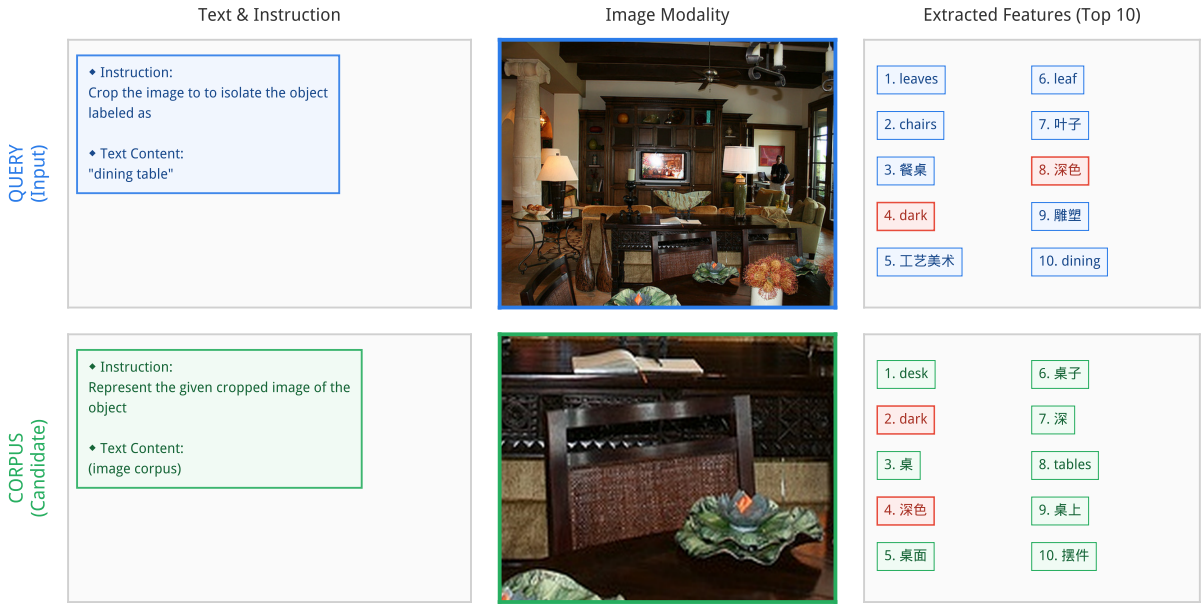


Figure 10: Case study of UEMBED on grounding (MSCOCO). We visualize the top-10 activated sparse tokens.